Equal Density Does Not Give a Constant Expected-Subgraph Ratio

A counterexample to TLMC 00000006182

Galaxy-0

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Abstract

The triangle and the four-cycle are strictly balanced graphs, both of density one. Their expected numbers of unlabelled, not necessarily induced copies in $G(n,p)$ have ratio $4/[3(n-3)p]$. Already for $n=4$, the ratio is $16/3$ at $p=1/4$ and $8/3$ at $p=1/2$. This disproves the conjecture's unqualified constant-ratio clause. Lean 4.31.0 with only Std checks the actual graph subsets, strict balance, the complete 64-outcome sample space, symbolic expectation identities, and a contradiction valid over every characteristic-zero field. The common-threshold theorem and the narrower fixed-scale asymptotic statement are not contradicted.

1 Claim, interpretation, and exact scope

The official English statement ends by asserting that two strictly balanced subgraphs share a threshold exactly when their densities are equal, "in which case the expectation ratio is constant." The Chinese statement has the same unqualified constancy claim. We use the standard model $G(n,p)$: the vertex set is $\{0,\dots,n-1\}$, and each unordered pair is independently an edge with probability p. For a fixed graph H, let N_H count *unlabelled subgraphs* isomorphic to H. They need not be induced. Different edge subsets count as different copies, and every copy counts once, regardless of its automorphisms.

Our counterexample addresses an *exact* ratio claimed to be independent of the edge probability. In particular it refutes a universal constant conversion factor obtained solely from equal densities. It does not refute the ordinary threshold $p \asymp n^{-1}$ for cycles. It also does not refute a ratio that merely *converges* to a constant after fixing $p=c/n$ with one fixed $c>0$. Section 4 makes the distinction explicit. If that narrower asymptotic claim was intended, it requires an additional qualification absent from the source.

2 Strict balance and the general counting formula

Write $v(H)$ and $e(H)$ for the numbers of vertices and edges, and $d(H)=e(H)/v(H)$ for nonempty H. A graph is strictly balanced if every proper subgraph with at least one vertex has smaller density. A subgraph may discard both vertices and edges; isolated vertices are allowed.

Lemma 1. *For every $k\ge 3$, the cycle C_k is strictly balanced and has density one.*

Proof. It has k edges and k vertices. A proper subgraph cannot contain a cycle: any cycle contained in C_k must traverse all its edges and vertices. Thus every proper subgraph J is a forest. If $v(J)>0$, then $e(J)\le v(J)-1<v(J)$, so $d(J)<1=d(C_k)$. □

Lemma 2. *For $n\ge k\ge 3$,*

$$\mathbb{E}N_{C_k}=\frac{(n)_k}{2k}p^k,\qquad (n)_k=n(n-1)\cdots(n-k+1).$$

Proof. There are $(n)_k$ ordered injective placements of the cyclically ordered vertices into $[n]$. Each unlabelled cycle has exactly $2k$ descriptions: a choice of starting vertex and one of the two traversal directions. Each candidate copy requires k distinct edges, hence is present with probability p^k by independence. Extra edges have no effect because the copies need not be induced. Sum the indicator expectations over the $(n)_k/(2k)$ candidate copies. □

In particular, for all $n \geq 4$ and $p > 0$,

$$\mathbb{E}N_{C_3} = \binom{n}{3}p^3, \qquad \mathbb{E}N_{C_4} = 3\binom{n}{4}p^4, \tag{1}$$

$$\frac{\mathbb{E}N_{C_3}}{\mathbb{E}N_{C_4}} = \frac{4}{3(n-3)p}. \tag{2}$$

The factor 3 in the denominator of (2) is essential: the automorphism factors for C_3 and C_4 are 6 and 8.

3 A finite counterexample with interior probabilities

Theorem 1. *The equally dense, strictly balanced graphs C_3 and C_4 have no constant expectation ratio in $G(4,p)$ as p varies in $(0,1)$.*

Proof. Equation (1) gives $\mathbb{E}N_{C_3} = 4p^3$ and $\mathbb{E}N_{C_4} = 3p^4$. Exact values are

p	$\mathbb{E}N_{C_3}$	$\mathbb{E}N_{C_4}$	$\mathbb{E}N_{C_3}/\mathbb{E}N_{C_4}$
1/4	1/16	3/256	16/3
1/2	1/2	3/16	8/3

Both denominators are positive, and both probabilities are strictly between zero and one. Equivalently, the two cross products are

$$\frac{1}{16}\cdot\frac{3}{16} = \frac{3}{256} \quad \neq \quad \frac{1}{2}\cdot\frac{3}{256} = \frac{3}{512}.$$

Thus no real number a can simultaneously satisfy $\mathbb{E}N_{C_3} = a\mathbb{E}N_{C_4}$ at both probabilities. □

4 What happens at the common threshold

At $p = c/n$, where $c > 0$ is fixed, the exact ratio is

$$R_n = \frac{4n}{3c(n-3)} = \frac{4}{3c} + \frac{4}{c(n-3)}.$$

Consequently $R_n \to 4/(3c)$. This correct, narrower asymptotic constancy is fully compatible with the counterexample. The limiting constant depends on c; even along a fixed scale the finite-n ratio is not exactly constant. For comparison, $p = n^{-1/2}$ makes the ratio tend to zero, while $p = n^{-2}$ makes it tend to infinity. These conclusions follow immediately from the exact formula. None requires denying the standard common threshold.

5 Formal model and audit boundary

The sole mathematical source file is `Main.lean`, importing `Std`. The formal model is finite but uses actual graphs:

- Vertices are `Fin 4`. The six edges are the actual unordered pairs $01, 02, 03, 12, 13, 23$, in that order. A graph consists of a four-bit vertex subset and a six-bit edge subset. Well-formedness requires every edge endpoint to be present. Subgraphs are inclusion of both subsets.

- Strict balance is checked against *all* proper subgraphs with nonempty vertex set, using the equivalent cross-multiplied density inequality. The chosen graphs have vertex/edge masks $(7, 11)$ and $(15, 45)$.

- A three- or four-vertex cycle is defined as a well-formed graph with the specified number of vertices and degree two at every vertex. Exhaustive classification gives exactly four triangles and three four-cycles. A separate theorem proves connectivity using actual walks. The representative lists have no duplicates, so filtering them by subgraph inclusion counts copies without automorphism multiplicity.

- Outcomes are all 64 edge masks, with all four vertices retained. Every Boolean assignment to the six edges occurs exactly once. The probability of an outcome with m edges is $p^m(1-p)^{6-m}$; its equality with the product of its six individual Bernoulli factors is proved. Total mass is one, and for rational $0 \le p \le 1$ every mass is nonnegative.

- Expectation is the finite weighted sum of the actual copy count over these outcomes. Kernel-checked symbolic identities give $4p^3$ and $3p^4$. They hold in every field. The final characteristic-zero-field theorem rules out a common proportionality constant at $p = 1/4, 1/2$, so the contradiction applies to real constants, not only rational ones.

The complete disproof in Theorem 1 is formalized. The arbitrary-n enumeration proof in Lemma 2 and the asymptotic discussion are ordinary mathematical proofs in this report; they are not claimed as Lean theorems. They provide context and are unnecessary for the finite counterexample. No external probability library, graph library, assumed expectation formula, custom axiom, `sorry`, `native_decide`, or kernel-bypassing evaluation is used.

6 Reproduction and source

Use Lean 4.31.0, commit `68218e876d2a38b1985b8590fff244a83c321783`. There are no Lake dependencies. From the directory containing this report, run `./VERIFY.sh`. It performs a Lake build, re-elaboration with warnings treated as errors, independent `leanchecker` kernel replay of both modules, axiom reports, and a Python standard-library check. The Python check independently constructs cycles as sets of actual edges, enumerates all outcomes for $n = 4, 5$, multiplies Bernoulli factors edge by edge, and computes exact rational expectations. It is an additional check, not a source of Lean proof authority. The proof audits report only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`, as applicable.

The source is TLMC conjecture 00000006182, with verified source blob:

`920179a96b1058c56db58613db3647e61f614cd0`.

As checked on 17 October 2026, its metadata has neither proven nor disproven set; the 1,024-item combined historical pull-request snapshot contains no prior submission referring to this conjecture. Eligibility and reproduction logs accompany the source package.